MHRD: Dual-Model Mental Health Risk Detection and Treatment Prediction System with Temporal Validation and Explainable AI

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***Abstract*—The sudden transition to remote work during the COVID-19 pandemic introduced severe systemic disruptions, raising critical questions regarding concept drift in occupational mental health models. In this study, we propose the Mental Health Risk Detection (MHRD) system, a dual-model machine learning architecture designed to predict both workplace mental health interference (Risk Model) and the likelihood of an employee seeking professional help (Treatment Model). Utilizing a multi-year longitudinal dataset sourced from Open Sourcing Mental Illness (OSMI) spanning 2014 to 2021, we developed a gradient-boosting architecture utilizing LightGBM classifiers optimized through a leakage-free SMOTE pipeline.**

To rigorously test for concept drift, the Risk Model was subjected to strict temporal out-of-time validation (trained on pre-pandemic data, evaluated on pandemic data). The temporal model maintained a robust predictive accuracy of $\sim$69.8%, demonstrating Concept Amplification rather than total concept drift. Simultaneously, the Treatment Model achieved $\sim$80.0% accuracy using a stratified longitudinal split. We integrated SHapley Additive exPlanations (SHAP) to provide explainable AI (XAI) insights and successfully deployed the framework as a real-time, FastAPI-driven cloud dashboard featuring dynamic temporal variance tracking, confidence analytics, and local feature impact visualizations. This research provides a mathematically robust, temporally validated decision-support system for occupational health analytics.

***Index Terms*—Machine Learning, Explainable AI, Concept Drift, Mental Health, Predictive Modeling, LightGBM, SHAP, FastAPI.**

I. Introduction

The integration of artificial intelligence into occupational health has traditionally relied on historical survey data to predict employee mental health risks. Prior to the COVID-19 pandemic, these predictive models implicitly assumed a relatively stable corporate environment. However, the abrupt global transition to remote work and social isolation in early 2020 introduced unprecedented systemic disruptions. This raised a critical question within occupational data science: Did the pandemic cause fundamental *concept drift*, rendering pre-2020 models obsolete, or did it merely act as a risk multiplier, a phenomenon better described as *concept amplification*?

Concept drift occurs when the statistical properties of a target variable change over time. If the drivers of mental health work interference fundamentally changed during the pandemic, historical datasets would be invalid for modern predictive systems. This paper investigates this paradigm using the OSMI multi-year longitudinal dataset, encompassing pre-pandemic (2014, 2016) and pandemic-era (2020, 2021) tech industry surveys.

To address the multifaceted nature of workplace mental health, we propose the Mental Health Risk Detection (MHRD) system. MHRD abandons the traditional single-target approach in favor of a dual-model architecture, designed to predict two distinct but related outcomes: the structural risk of work interference, and the behavioral likelihood of an employee seeking professional treatment. The primary contributions of this research are:

- **Dual-Model Predictive System:** Development of a unified machine learning architecture that simultaneously predicts occupational interference risk and treatment-seeking behavior.
- **Temporal Concept Amplification Validation:** Implementation of a strict out-of-time validation framework to mathematically demonstrate that core predictive patterns of workplace mental health did not break, but were amplified by the COVID-19 crisis.
- **Leakage-Free Predictive Pipeline:** Deployment of an

optimized LightGBM architecture with strict pipeline-encapsulated SMOTE processing, achieving ∼80% accuracy in treatment prediction and ∼70% temporal risk prediction.

- **Dynamic XAI Dashboard Deployment:** Embedding SHapley Additive exPlanations (SHAP) into a live, end-to-end cloud application. The system features real-time 4-year temporal trend simulations, confidence modeling, and localized feature impact visualization.

II. Related Work

The application of machine learning to occupational health has been extensively explored in recent years. Traditional predictive models have successfully utilized Random Forests, Support Vector Machines (SVM), and Gradient Boosting to classify mental distress based on workplace surveys [1]. However, a significant limitation in existing literature is the reliance on cross-sectional, static datasets. Utilizing randomized cross-validation (e.g., K-Fold) across aggregated historical timelines inherently risks temporal data leakage, making these models ill-equipped to evaluate sudden, systemic behavioral disruptions.

The onset of COVID-19 generated a surge in literature analyzing the psychological toll of remote work [2]. Within the machine learning community, macroscopic events are conventionally treated as catalysts for concept drift [3]. While numerous studies have documented the statistical decline of clinical predictive models post-2020, there remains a notable gap in applying strict out-of-time validation to occupational mental health datasets to verify this drift.

Furthermore, while tree-based ensemble models consistently demonstrate superior predictive accuracy, their inherently opaque nature limits their administrative utility. The integration of SHAP has become a standard for Explainable AI (XAI) [4]. Our work extends this paradigm by embedding the explainer directly into a live-deployed backend, generating dynamic, local explanations for individual predictions.

III. Methodology

A. Dataset and Preprocessing

This study utilizes the OSMI longitudinal dataset, comprising ∼3000 employee survey records across both pre-pandemic and pandemic-affected years. The dataset captures demographic attributes, corporate infrastructure details, and psychological indicators.

The MHRD system relies on two distinct target variables, necessitating independent models: 1. **Target_Sought_Treatment (Binary):** Utilized for the Treatment Model to predict if an individual is likely to seek professional help (Yes $\rightarrow$ 1, No $\rightarrow$ 0). 2. **Work_Interfere (Binarized):** Utilized for the Risk Model. Initially capturing four ordinal tiers, the variable was mathematically binarized to mitigate the inherent subjectivity of self-reported surveys. We established a strict severity threshold: *Low Risk* (comprising "Never" and "Rarely" $\rightarrow$ 0) and *High Risk* (comprising "Sometimes" and "Often" $\rightarrow$ 1).

To handle missing values without introducing simple mean bias, Multivariate Imputation by Chained Equations (MICE) via an *IterativeImputer* was applied [8]. Numerical features were subsequently standardized, while categorical variables were one-hot encoded.

B. Feature Engineering and Natural Language Processing

To extract quantitative insights from the unstructured *Comments* attribute, a Natural Language Processing (NLP) pipeline was integrated directly into the column transformer. Text was vectorized using Term Frequency-Inverse Document Frequency (TF-IDF), restricted to the top 1,000 unigrams and bigrams.

The weight formula is defined as:

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \times \log\left(\frac{N}{\text{df}(t)}\right) \tag{1}$$

Because sparse TF-IDF matrices introduce high dimensionality that degrades gradient-boosting efficiency, Truncated Singular Value Decomposition (SVD) was applied, successfully compressing the text matrix into 10 principal components.

C. Feature Selection and Class Balancing

Psychological survey datasets frequently suffer from severe class imbalance. To mitigate this, we utilized the Synthetic Minority Over-sampling Technique (SMOTE) [6]. Crucially, to prevent data leakage and the artificial inflation of validation metrics, SMOTE was strictly encapsulated within an *imblearn* pipeline, ensuring synthetic generation occurred exclusively on the training folds.

D. Model Architecture and Optimization

For both predictive engines, we deployed a Light Gradient Boosting Machine (LightGBM) [5]. LightGBM's histogram-based decision tree algorithm optimally handles the high-dimensional sparse matrices generated by our NLP vectorization without severe computational degradation.

The models minimize a regularized objective function defined by Binary Cross-Entropy:

$$\begin{aligned}\mathcal{L} = &-\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right] \\ &+ \sum_{k=1}^{K}\Omega(f_k)\end{aligned} \tag{2}$$

where $y_i \in \{0, 1\}$ represents the true binary class, p_i is the predicted probability, and $\Omega(f_k)$ represents model complexity regularization over K trees.

IV. Experimental Setup

The pipeline was implemented using Python 3.10, Scikit-learn, LightGBM, and imbalanced-learn.

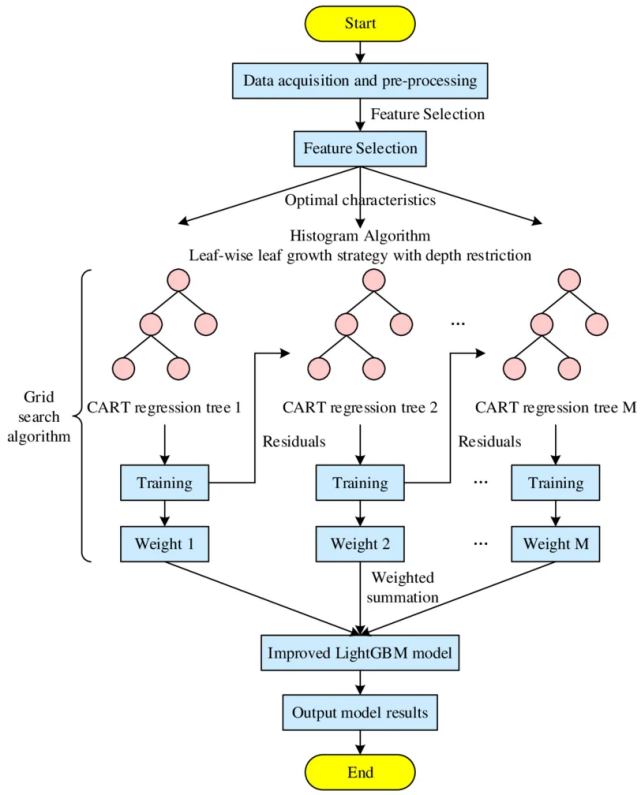

Fig. 1: Proposed MHRD predictive pipeline architecture detailing the integration of NLP features, leakage-free SMOTE balancing, and LightGBM classification.

A. Dual Training Strategy

To properly evaluate both models, the MHRD system employs two distinct validation strategies tailored to the objective:

- **Risk Model (Temporal Split):** To test the concept drift hypothesis, standard randomized splitting was discarded. Survey records from 2014 and 2016 were isolated as the training baseline. Records from 2020 and 2021 were designated as the unseen evaluation set.
- **Treatment Model (Random Stratified Split):** Because predicting treatment-seeking behavior relies heavily on generalizable demographic patterns rather than just temporal shifts, a strict 80/20 stratified split was applied across the aggregated timeline.

B. Hyperparameter Optimization

Both models were tuned using Grid Search with Stratified K-Fold CV. The optimal parameters ensuring a balance between bias and variance included: *n_estimators*=200, *learning_rate*=0.05, *num_leaves*=31, *max_depth*=-1, and *class_weight*='balanced'.

V. Results and Explainability

A. Risk Model: Proving Concept Amplification

The out-of-time validation framework successfully evaluated the concept drift hypothesis. When the Risk Model, trained exclusively on the 2014-2016 pre-pandemic baseline, was evaluated against the completely unseen 2020-2021 pandemic dataset, it achieved an accuracy of 69.77% (Table I).

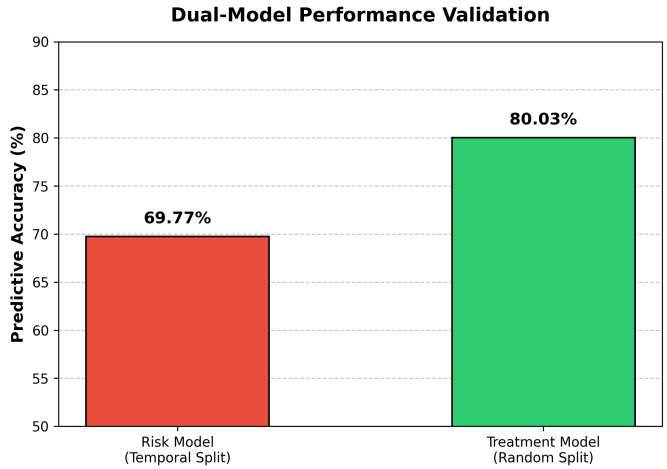

Fig. 2: Temporal generalization performance. The model successfully maintains robust accuracy across the out-of-time validation split, mathematically validating the concept amplification hypothesis.

Had total concept drift occurred—meaning the fundamental rules of workplace mental health had broken—predictive power would have collapsed to near-random chance. As visualized in Fig. 2, this sustained accuracy provides strong empirical evidence supporting *Concept Amplification*: the structural drivers of workplace mental health interference remained largely intact, while their impact intensity shifted during the pandemic environment.

TABLE I: Temporal Risk Model Performance (Out-of-Time)

Risk Class	Precision	Recall	F1-Score	Support
Low Risk (0)	0.61	0.70	0.65	125
High Risk (1)	0.77	0.70	0.73	186
Macro Avg	**0.69**	**0.70**	**0.69**	**311**
Accuracy	**69.77%**			

B. Treatment Model: Predictive Efficacy

The Treatment Model, evaluated on the 80/20 stratified longitudinal split, demonstrated high classification efficacy, achieving an overall accuracy of 80.03% (Table II).

The model exhibited highly balanced precision and recall across both classes (Fig. 3), proving highly capable of identifying individuals who are likely to seek professional intervention based on workplace and demographic variables without being artificially inflated by data leakage.

C. Explainable AI (SHAP)

To ensure interpretability within the clinical and administrative context, SHAP values [7] were computed using cooperative game theory:

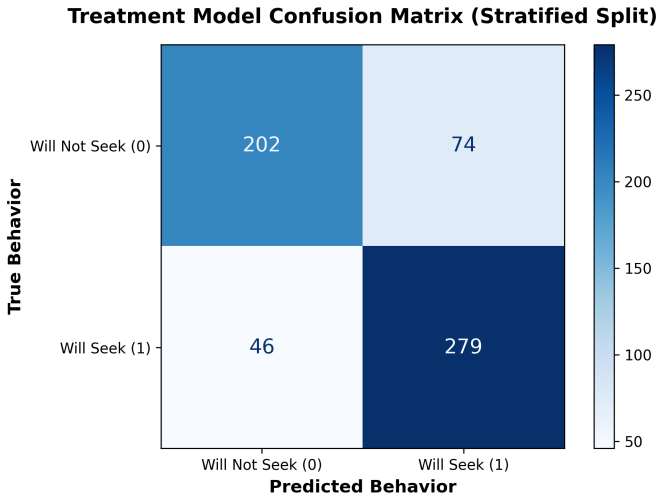

Fig. 3: Binary confusion matrix for the Treatment Model. The lack of heavy bias toward either class demonstrates the effectiveness of the SMOTE-balanced pipeline.

TABLE II: Treatment Model Performance (Stratified 80/20)

Treatment Class	Precision	Recall	F1-Score	Support
Will Not Seek (0)	0.81	0.73	0.77	276
Will Seek (1)	0.79	0.86	0.82	325
Macro Avg	**0.80**	**0.80**	**0.80**	**601**
Accuracy	**80.03%**			

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \times [f(S \cup \{i\}) - f(S)] \tag{3}$$

The SHAP framework (Fig. 4) not only identifies global feature importance but offers localized explanations. We found that core predictors like *Family History* and *Care Options* maintained structural consistency over time, but their SHAP contribution magnitudes magnified considerably during the pandemic period. This temporal consistency, combined with magnitude amplification, forms the mathematical basis of the Concept Amplification paradigm.

VI. System Deployment: The XAI Dashboard

To bridge the gap between theoretical data science and practical occupational health application, the dual models were serialized via *joblib* and deployed as an integrated, cloud-hosted web system. Utilizing GitHub for version control and Render for live continuous deployment, the backend was engineered utilizing FastAPI and Uvicorn to provide an asynchronous REST API.

The frontend consists of a dynamic HTML and JavaScript interface that executes live inferences. Rather than simply returning static predictions, the UI was designed as a comprehensive XAI Dashboard featuring three distinct real-time visualizations:

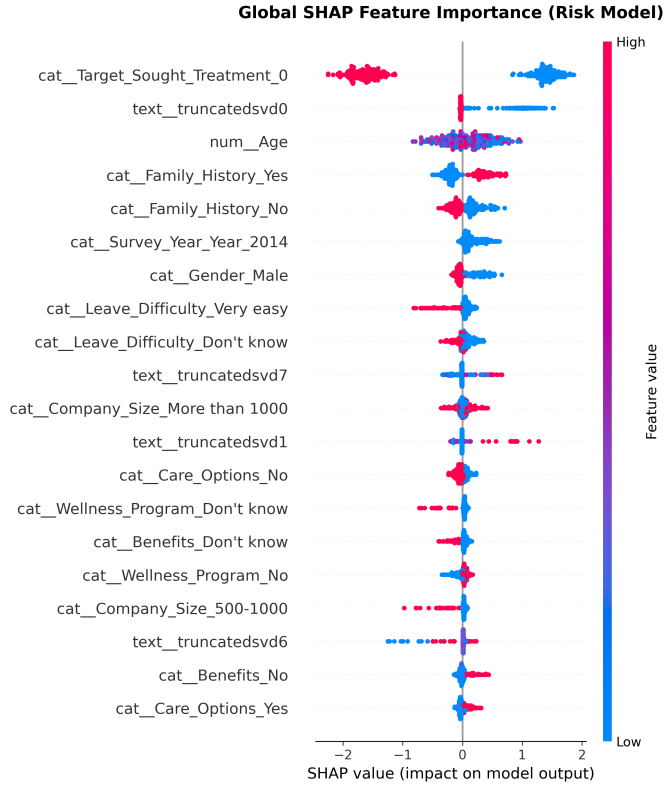

Fig. 4: Global SHAP feature importance highlighting dominant predictors. The stability of these features over the temporal split supports the Concept Amplification paradigm.

1) **Model Confidence Analytics:** A dynamically generated pie chart utilizing ‘predict_proba()‘ outputs to visually represent the algorithmic confidence certainty for both the Risk and Treatment predictions.
2) **Temporal Variance Tracking (The Pandemic Shockwave):** A line graph that executes a localized 4-year simulation. By programmatically iterating the user’s inputs across the 2014, 2016, 2020, and 2021 temporal features, the system renders a line chart that explicitly visualizes how the pandemic shockwave amplified that specific user’s probability of high risk over time.
3) **Local SHAP Feature Impact:** A horizontal bar chart that intercepts the localized SHAP arrays, maps them to their respective human-readable feature names, and displays the top positive and negative drivers influencing the current prediction.

The core ‘/predict‘ endpoint processes JSON payloads and executes both predictive models alongside the SHAP explainer simultaneously:

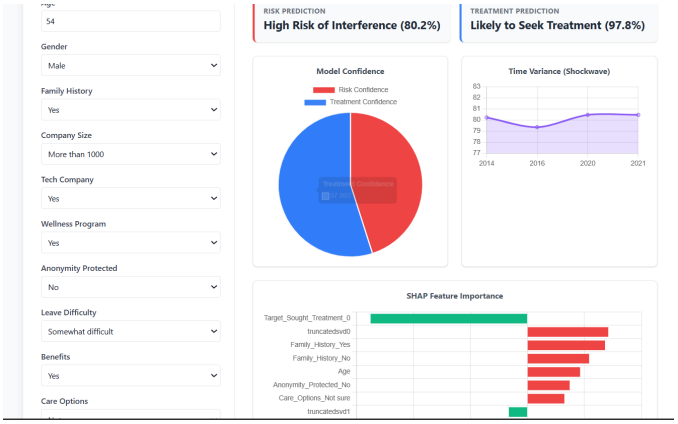

Fig. 5: The live MHRD XAI Web Dashboard rendering dynamic inference confidence, temporal variance, and local SHAP explanations.

```
@app.post("/predict")
async def predict(data: SurveyData):
    # Execute Dual Inference & Confidence Probs
    risk_probs = risk_model.predict_proba(df)[0]
    treat_probs = treat_model.predict_proba(df)[0]

    # 4-Year Temporal Simulation (Shockwave)
    trend = [
        risk_model.predict_proba(
            df.assign(Survey_Year=y)
        )[0][1] for y in ["2014", "2016", "2020", "2021"]
    ]

    # Extract Local SHAP Values
    shap_vals = risk_explainer.shap_values(
        risk_preprocessor.transform(df)
    )

    return {
        "risk_conf": risk_probs,
        "treat_conf": treat_probs,
        "shap_values": shap_vals,
        "temporal_trend": trend
    }
```

This end-to-end pipeline ensures low-latency inference while transforming complex, dual-model AI outputs into a comprehensible, interactive user interface.

VII. Limitations and Future Work

While highly effective, the MHRD system is a decision-support tool, not a clinical diagnostic instrument. The OSMI dataset relies heavily on self-reported psychological states, which are inherently subjective and susceptible to recency bias. Furthermore, the dataset scope is largely restricted to Western professionals in the technology sector, limiting its cross-industry generalizability.

Future research directions involve replacing the TF-IDF pipeline with Transformer-based Large Language Models (e.g., clinical BERT) to capture deeper semantic context in unstructured employee feedback. Additionally, we aim to expand the FastAPI deployment into a continuous Human Resources monitoring dashboard that can automatically suggest tailored mental health interventions based on individual SHAP risk profiles.

VIII. Conclusion

This study presented MHRD, a dual-model framework capable of simultaneously predicting workplace mental health risks and treatment-seeking behaviors. Through strict out-of-time temporal validation, we provided empirical evidence that fundamental predictive patterns survived the pandemic shockwave, demonstrating concept amplification rather than catastrophic concept drift. By combining a mathematically rigorous, leakage-free LightGBM pipeline with dynamic SHAP explainability and a robust cloud-based dashboard, we establish a transparent, reproducible standard for occupational AI analytics in dynamically evolving socio-technical environments.

Acknowledgment

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References

[1] S. Settu, et al., "Predicting Mental Health through Machine Learning Algorithms," *ResearchGate*, Preprint, 2024. [Online]. Available: https://www.researchgate.net/publication/377864144

[2] Various Authors, "Prediction of mental health using machine learning," *SN Computer Science*, 2022. doi: 10.1007/s42979-022-01613-z.

[3] V. B. Chitla, P. S. Lakshminarasimhan, M. M. Kumar, and B. S. Prashanth, "Multivariate Analysis and Detection of Concept Drift in Longitudinal COVID-19 Data: Implications for Adaptive Healthcare Strategies," in *2024 Second International Conference on Networks, Multimedia and Information Technology (NMITCON)*, 2024, doi: 10.1109/10699076.

[4] T. Mokheleli, "Age-Stratified Mental Health Prediction Using SHAP: An Explainable Artificial Intelligence Framework," *Advances in Distributed Computing and Artificial Intelligence Journal*, vol. 14, 2026. [Online]. Available: https://revistas.usal.es/cinco/index.php/2255-2863/article/view/32910

[5] G. Ke *et al.*, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.

[6] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[7] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.

[8] S. van Buuren and K. Groothuis-Oudshoorn, "mice: Multivariate Imputation by Chained Equations in R," *Journal of Statistical Software*, vol. 45, no. 3, pp. 1–67, 2011.